

Question 1

```
5 --Question 1 to pull transactions occurred in 2023
6
7 SELECT *
8 FROM 'sales-retail-493417.RetailSales.Retail Sales'
9 WHERE EXTRACT(YEAR FROM Date) = 2023;
```

✓ This script will process 145.38 KB when run.
Using on-demand processing quota

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Job information Results Visualization JSON Execution details Execution graph

ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit
387	2023-06-04	CUST387	Male	44	Beauty	1	30
396	2023-02-23	CUST396	Female	55	Beauty	1	30
670	2023-10-05	CUST670	Male	27	Beauty	1	30
733	2023-08-29	CUST733	Male	34	Beauty	1	30
747	2023-11-15	CUST747	Male	23	Beauty	1	30
749	2023-05-03	CUST749	Male	42	Beauty	1	30

Question 2

```
10
11 ---Question 2 Display transactions where the Total_Amount is more than the average Total_Amount of the entire dataset.
12 SELECT *
13 FROM 'sales-retail-493417.RetailSales.Retail Sales'
14 WHERE 'Total Amount' > (SELECT AVG('Total Amount')
15 FROM 'sales-retail-493417.RetailSales.Retail Sales');
16
17 -----
18
```

✓ This query will process 72.69 KB when run.
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Job information Results Visualization JSON Execution details Execution graph

Row	Transaction ID	Date	Customer ID	Gender	Age	Product Cat...	Quant...	Price per Unit	Total Amount
1	21	2023-01-14	CUST021	Female	50	Beauty	1	500	500
2	28	2023-04-23	CUST028	Female	43	Beauty	1	500	500
3	128	2023-07-05	CUST128	Male	25	Beauty	1	500	500
4	220	2023-03-03	CUST220	Male	64	Beauty	1	500	500
5	238	2023-01-17	CUST238	Female	39	Beauty	1	500	500
6	364	2023-08-23	CUST364	Female	19	Beauty	1	500	500
7	408	2023-04-15	CUST408	Female	64	Beauty	1	500	500

Question 3

```

17 -----
18 ---Question 3 Display total Revenue
19
20 SELECT SUM(`Total Amount`) AS Total_Revenue
21 FROM `sales-retail-493417.RetailSales.Retail Sales`;

```

 This script will process 225.88 KB when run.

Using on-demand processing quota

Query results

Job information	Results	Visualization	JSON	Execution detail
Row	Total_Revenue ▾			
1	456000			

Question 4

```

23 ---Question 4 : Display all distinct Product Categories in the dataset.
24
25 SELECT DISTINCT `Product Category`
26 FROM `sales-retail-493417.RetailSales.Retail Sales`;
27

```

 Query completed

Query results

 Chat

Job information	Results	Visualization	JSON	Execution details	Execution graph
Row	f0_ ▾				
1	Product Category				

Question 5

```

28 ---Question 5 : For each Product Category, calculate the total quantity sold.GROUP BY
29
30 SELECT 'Product Category', SUM(Quantity) AS Total_Quantity
31 FROM `sales-retail-493417.RetailSales.Retail Sales`
32 GROUP BY 'Product Category';
33

```

This script will process 243.86 KB when run.

Using on-demand processing quota

Query results

Job information	Results	Visualization	JSON	Execution details	Execution graph
Row	Product Category	Total_Quantity			
1	Beauty	771			
2	Clothing	894			
3	Electronics	849			

Question 6

```

34 ----Question 6 Create a column called Age_Group that classifies customers as:- 'Youth' (<30)
35
36 SELECT
37     'Customer ID',
38     Age,
39     CASE
40         WHEN Age < 30 THEN 'Youth'
41         WHEN Age BETWEEN 30 AND 59 THEN 'Adult'
42         ELSE 'Senior'
43     END AS Age_Group
44 FROM `sales-retail-493417.RetailSales.Retail Sales`;

```

Query completed

Using on-demand processing quota

Query results

Job information	Results	Visualization	JSON	Execution details	Execution graph
Row	Customer ID	Age	Age_Group		
1	CUST191	64	Senior		
2	CUST204	39	Adult		
3	CUST230	54	Adult		
4	CUST232	43	Adult		
5	CUST309	26	Youth		
6	CUST310	28	Youth		
7	CUST363	64	Senior		
8	CUST371	20	Youth		
9	CUST397	30	Adult		
10	CUST454	46	Adult		

Question 7

```
6
7 --- QUESTION 7: For each Gender, count how many high-value transactions occurred (where Total Amount > 500). Expected Output: Gender,
  High_Value_Transactions
8 SELECT
9   Gender,
10  COUNTIF('Total Amount' > 500) AS High_Value_Transactions
11 FROM 'sales-retail-493417.RetailSales.Retail Sales'
12 GROUP BY Gender;
```

This script will process 275.13 KB when run.

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Query results [Chat](#) [Save results](#)

Job information **Results** Visualization JSON Execution details Execution graph

	Gender	High_Value_Trans...
1	Male	144
2	Female	155

Question 8

```
54 --Question 8 For each Product Category, show only those categories where the total revenue exceeds 5,000.
55
56 SELECT
57   'Product Category',
58   SUM('Total Amount') AS 'Total Revenue'
59 FROM 'sales-retail-493417.RetailSales.Retail Sales'
60 GROUP BY 'Product Category'
61 HAVING SUM('Total Amount') > 5000;
62
```

Query completed

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Job information **Results** Visualization JSON Execution details Execution graph

Row	Product Category	Total Revenue
1	Beauty	143515
2	Clothing	155580
3	Electronics	156905

Question 9

```

68 SELECT
69     'Transaction ID',
70     'Price per Unit',
71     CASE
72         WHEN 'Price per Unit' < 50 THEN 'Cheap'
73         WHEN 'Price per Unit' BETWEEN 50 AND 200 THEN 'Moderate'
74         WHEN 'Price per Unit' > 200 THEN 'Expensive'
75     END AS Unit_Cost_Category
76 FROM 'sales-retail-493417.RetailSales.Retail Sales';
77

```

✓ This script will process 308.73 KB when run.
Using on-demand processing quota

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Job information	Results	Visualization	JSON	Execution details	Execution graph
Row	Transaction ID	Price per Unit	Unit_Cost_Category		
36	27	25	Cheap		
37	40	50	Moderate		
38	102	25	Cheap		
39	120	50	Moderate		
40	172	25	Cheap		
41	189	50	Moderate		
42	207	25	Cheap		
43	273	50	Moderate		
44	349	50	Moderate		

Question 10

```

78
79 --QUESTION 10 Display all transactions from customers aged 40 or older and add a column Spending_Level showing: 'High' if Total Amount > 1000 'Low'
otherwise
80 SELECT
81     'Customer ID',
82     'Age',
83     'Total Amount',
84     CASE
85         WHEN 'Total Amount' > 1000 THEN 'High'
86         ELSE 'Low'
87     END AS Spending_Level
88 FROM 'sales-retail-493417.RetailSales.Retail Sales'

```

✓ Query completed
Using on-demand processing quota

Query results [Chat](#) [Save results](#) [Open in](#)

Job information	Results	Visualization	JSON	Execution details	Execution graph
Row	Customer ID	Age	Total Amount	Spending_Level	
3	CUST232	43	25	Low	
4	CUST363	64	25	Low	
5	CUST454	46	25	Low	
6	CUST512	57	25	Low	
7	CUST791	51	25	Low	
8	CUST825	46	25	Low	
9	CUST855	54	25	Low	
10	CUST967	62	25	Low	
11	CUST006	45	30	Low	
12	CUST057	63	30	Low	